

CBSE 2027 Maths — Predicted Paper (Set 10 of 10)

Based on a comprehensive analysis of the latest CBSE blueprint, repeating previous year question (PYQ) trends, and the official syllabus marks distribution (Calculus 35, Vectors & 3D 14, Algebra 10, Probability 8, Relations & Functions 8, LPP 5), here is the 10th and final set of our predicted high-probability Class 12 Maths board paper.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3$. Determine whether the function f is one-one, onto, bijective, or neither. * **Chapter:** Relations and Functions * **Probability Tag: High**

Q2. Find the exact principal value of $\cos^{-1}\left(-\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If A is a 3×3 matrix such that $A^T = -A$, what is the trace of matrix A (the sum of its principal diagonal elements)? * **Chapter:** Matrices * **Probability Tag: Medium**

Q4. The area of a triangle is 9 sq. units and its vertices are $(-3, 0)$, $(3, 0)$, and $(0, k)$. Find the positive value of k using determinants. * **Chapter:** Determinants * **Probability Tag: High**

Q5. What is the derivative of $\log_{10} x$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The volume of a cube is increasing. Find the rate of change of its volume with respect to its edge length a when the edge is 3 cm. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate the indefinite integral: $\int \frac{1-\sin x}{\cos^2 x} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q8. Find the area of the region bounded by the curve $y = \sqrt{x}$, the x-axis, and the vertical lines $x = 1$ and $x = 4$ in the first quadrant. * **Chapter:** Application of Integrals * **Probability Tag: Medium**

Q9. State the order of the differential equation: $\frac{d^4 y}{dx^4} + \sin\left(\frac{d^2 y}{dx^2}\right) = 0$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q10. Find the scalar projection of the vector $\vec{a} = \hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $\vec{b} = 7\hat{i} - \hat{j} + 8\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. Find the value of λ if the vectors $\vec{p} = 2\hat{i} + \lambda\hat{j} + 3\hat{k}$ and $\vec{q} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ are strictly perpendicular to each other. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q12. Find the direction cosines of the straight line passing through the origin and the point $(1, 2, 3)$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q13. Write the Cartesian equation of the plane that makes intercepts of 2, 3, and 4 on the x, y, and z axes respectively. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. In an LPP, the feasible region has corner points $(0, 0)$, $(0, 8)$, $(4, 10)$, $(6, 8)$, and $(6, 0)$. What is the maximum value of the objective function $Z = 3x + 9y$? * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. Two standard fair dice are thrown simultaneously. What is the probability of getting a sum strictly greater than 10? * **Chapter:** Probability * **Probability Tag: High**

Q16. Evaluate the definite integral using integration by parts: $\int_0^1 xe^x dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. Find the general solution of the differential equation: $\frac{dy}{dx} = \frac{x^2}{y^2}$. * **Chapter:** Differential Equations * **Probability Tag: Medium**

Q18. A family has exactly 3 children. Assuming that boys and girls are equally likely, find the probability that the family has exactly 2 girls given that the first child is a girl. * **Chapter:** Probability * **Probability Tag: Medium**

Q19. (Assertion-Reason): Assertion (A): If A is a symmetric matrix and is invertible, then its inverse A^{-1} is also a symmetric matrix. **Reason (R):** For any invertible square matrix A , $(A^{-1})^T = (A^T)^{-1}$. * **Chapter:** Matrices * **Probability Tag: High**

Q20. (Assertion-Reason): Assertion (A): $\int_{-\pi/2}^{\pi/2} \sin^3 x dx = 0$. **Reason (R):** For any continuous odd function $f(x)$, the definite integral $\int_{-a}^a f(x) dx = 0$. * **Chapter:** Integrals * **Probability Tag: Very High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Solve the following equation for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. If a curve is represented parametrically by $x = a \cos^3 \theta$ and $y = a \sin^3 \theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q23. Find a unit vector which is perpendicular to both the vectors $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{i} + \hat{j} - 2\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q24. Find the integrating factor and the general solution of the linear differential equation: $\frac{dy}{dx} + y = e^{-x}$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. A random variable X has the probability distribution $P(X = x) = cx$ for $x = 1, 2, 3, 4$, and $P(X = x) = 0$ otherwise. Find the value of constant c and calculate the Mean (Expected Value) of X . * **Chapter:** Probability * **Probability Tag: Medium**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the function $f : [-1, 1] \rightarrow \mathbb{R}$ given by $f(x) = \frac{x}{x+2}$ is one-one. Also, find the inverse of the function $f : [-1, 1] \rightarrow \text{Range}(f)$. * **Chapter:** Relations and Functions *

Probability Tag: Very High

Q27. If $A = \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix}$, compute $A^2 - 5A + 4I$ (where I is the identity matrix of order 3). * **Chapter:** Matrices * **Probability Tag: High**

Q28. Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = x^{\cos x} + (\cos x)^x$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the indefinite integral using the method of completing the square and standard formulas: $\int \frac{3x+5}{x^2+3x-18} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the homogeneous differential equation: $x^2 \frac{dy}{dx} + y(x+y) = 0$, given that $y = 1$ when $x = 1$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. A pair of dice is thrown 4 times. If getting a sum of 9 on the dice is considered a "success," find the probability of getting at least 3 successes using the Binomial Distribution. * **Chapter:** Probability * **Probability Tag: High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the following definite integral using properties of definite integrals:

$$\int_0^{\pi/2} \frac{\sin^4 x}{\sin^4 x + \cos^4 x} dx$$
 * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. A tank with a rectangular base and rectangular sides, open at the top, is to be constructed so that its depth is **2 m** and its volume is **8 m³**. If the building of the tank costs ₹70 per square metre for the base and ₹45 per square metre for the sides, what is the cost of the least expensive tank? Use calculus to maximize/minimize. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Find the shortest distance between the two skew lines given in vector form:
 $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$ * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q35. Using the method of integration, sketch the region and find the exact area bounded by the curves $y = x^2$ and $y = |x|$. * **Chapter:** Application of Integrals * **Probability Tag: High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Factory Production) A company produces three products X, Y, and Z every day. Their total combined production on a certain day is **45 tons**. It is found that the production of the third product (Z) exceeds the production of the first product (X) by **8 tons**. Furthermore, the total production of the first product (X) and the third product (Z) is exactly twice the production of the second product (Y). (i) Formulate the above daily production scenario as a system of three linear equations. (1 Mark) (ii) Express the system of

equations in the matrix form $\mathbf{AX} = \mathbf{B}$. (1 Mark) (iii) Find the determinant of matrix $\mathbf{A}$ and use the matrix inverse method to find the exact production of each product X, Y, and Z. (2 Marks) * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q37. Case Study 2: Vector Algebra (Drone Surveillance) A group of students at a tech festival is controlling three surveillance drones hovering stationary in 3D space. Relative to a ground control origin, Drone A is positioned at coordinates mapped to the vector $2\hat{i} + 3\hat{j} + \hat{k}$, Drone B is at $-\hat{i} + 2\hat{j} + 4\hat{k}$, and Drone C is at $3\hat{i} + \hat{j} - 2\hat{k}$. (i) Find the position vector representing the displacement from Drone A to Drone B ($\vec{AB}$). (1 Mark) (ii) Calculate the exact linear distance between Drone A and Drone B. (1 Mark) (iii) Using the vector cross product, calculate the exact area of the geometric triangle formed by the positions of the three drones in the sky. (2 Marks) * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Testing & Bayes' Theorem) In answering a question on a highly competitive multiple-choice test, a student either knows the true answer or randomly guesses. Let $\frac{3}{4}$ be the historical probability that this particular student actually knows the answer, and $\frac{1}{4}$ be the probability that they simply guess. Assume that a student who guesses at the answer has a success probability of $\frac{1}{4}$ of getting it correct (since there are 4 options). A student who knows the answer gets it correct with a probability of 1. (i) Let $\mathbf{E}$ be the event of answering correctly and $\mathbf{K}$ be the event of knowing the answer. What is the conditional probability $P(\mathbf{E}|\mathbf{K})$? (1 Mark) (ii) Using the Law of Total Probability, find the overall probability that the student answers the question correctly. (1 Mark) (iii) Given the evidence that the student answered the question correctly, use Bayes' theorem to calculate the exact probability that the student genuinely knew the answer (and didn't just guess). (2 Marks) *

Chapter: Probability * **Probability Tag: Very High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.